

ON A CLASS OF PARTIALLY BALANCED INCOMPLETE BLOCK DESIGNS

R.C. BOSE*

Colorado State University, Fort Collins, Colorado, U.S.A.

M.S. SHRIKHANDE

Pahlavi University, Shiraz, Iran.

Received 4 April 1978; revised manuscript received 16 May 1978

Recommended by A. J. Hoffman

Abstract: Bose and Clatworthy (1955) showed that the parameters of a two-class balanced incomplete block design with $\lambda_1 = 1$, $\lambda_2 = 0$ and satisfying $r < k$ can be expressed in terms of just three parameters r, k, t . Later Bose (1963) showed that such a design is a partial geometry (r, k, t, c) . Bose, Shrikhande and Singhi (1976) have defined partial geometric designs (r, k, t, c) , which reduce to partial geometries when $c = 0$. In this note we prove that any two class partially balanced (PBIB) design with $r < k$, is a partial geometric design for suitably chosen r, k, t, c and express the parameters of the PBIB design in terms of r, k, t, c and λ_2 . We also show that such PBIB designs belong to the class of special partially balanced designs (SPBIB) studied by Bridges and Shrikhande (1974).

1. Introduction

Given a set $V = \{x_1, x_2, \dots, x_v\}$ of v points we have a two-class association scheme if there is a relation of association between any two distinct points satisfying the following conditions:

- (a) Any two points are either first associates or second associates.
- (b) Each point has n_i i th associates ($i = 1, 2$).
- (c) If two points are i th associates, then the number of treatments common to the j th associates of the first and k th associates of the second is p_{jk}^i and is independent of the pair of points with which we start. Also $p_{jk}^i = p_{kj}^i$.

If we assume the constancy of v, n_1, p_{11}^1 and p_{11}^2 then the constancy of $p_{12}^1, p_{21}^1, p_{22}^1, p_{12}^2, p_{21}^2, p_{22}^2$ follows and $p_{12}^1 = p_{21}^1, p_{12}^2 = p_{21}^2$ [Bose and Clatworthy (1955)].

The concept of an m -class association scheme was first introduced by Bose and Shimamoto (1952). We need only the special case $m = 2$ for the purposes of this paper.

* The research of this author was supported by AFOSR grant no. 77-3127.

A graph P (without loops or multiple edges) with v vertices is said to be strongly regular (Bose, (1963)] if

- (i) P is regular of degree n_1 .
- (ii) Any two vertices of P are simultaneously adjacent to p_{11}^1 or p_{11}^2 other vertices of P according as they are adjacent or non-adjacent.

We may identify the v points of a two-class association scheme with the v vertices of a graph P , and consider two vertices adjacent or non-adjacent according as they are first associates or second associates. It is clear that a strongly regular P with parameters

$$(v, n_1, p_{11}^1, p_{11}^2), \quad (1.1)$$

is isomorphic with a two-class association scheme with the same parameters. The adjacency matrix of the graph P will also be denoted by P .

If we have a two-class association scheme or equivalently a strongly regular graph P with parameters (1.1), then we get a partially balanced incomplete block (PBIB) design D based on P , if we can arrange the v vertices (points) into b subsets (blocks) such that

- (α) Each block contains k points (all different),
- (β) Each point is contained in r blocks,
- (γ) if any two points are i th associates ($i = 1, 2$) then they occur together in λ_i blocks. Then

$$(v, b, r, k, \lambda_1, \lambda_2) \quad (1.2)$$

are said to be the parameters of D . Note that v occurs both in (1.1) and (1.2). There are a number of well-known relations among these parameters [Bose and Nair (1939)].

PBIB designs were first studied by Bose and Nair (1939), though the concept of association schemes was not explicitly introduced in this paper. For further background on partially balanced designs and association schemes see Nair and Rao (1942); Bose and Connor (1952); Bose, Shrikhande and Bhattacharya (1953), Bose and Clatworthy (1955), Bose and Mesner (1959) and Dembowski (1968, pp. 281–291, 294, 303, 317).

Bose (1963) defined a partial geometry (r, k, t) as follows. We have a system of undefined points and lines together with an incidence relation satisfying the following axioms:

- A1: Any two points are incident with not more than one line.
- A2: Each point is incident with r lines.
- A3: Each line is incident with k points.
- A4: If the point P is not incident with the line l , there pass through P exactly t lines ($t \geq 1$) intersecting l .

By taking the points of the partial geometry as the points and the lines as the

blocks, we get a two-class PBIB design D with parameters

$$v = \frac{k}{t}[(r-1)(k-1) + t], \quad b = \frac{vr}{k}, \quad r, k, \lambda_1 = 1, \quad \lambda_2 = 0 \quad (1.3)$$

based on the strongly regular graph (association scheme) P with parameters

$$v, n_1 = r(k-1), \quad p_{11}^1 = (t-1)(r-1) + k - 2, \quad p_{11}^2 = rt \\ (1 \leq t \leq r, 1 \leq t \leq k). \quad (1.4)$$

Thus a partial geometry (r, k, t) is isomorphic to a PBIB design $(v, b, r, k, \lambda_1, \lambda_2 = 0)$ based on the graph of the geometry. However, an arbitrary PBIB design based on a strongly regular graph is not necessarily a partial geometry. Bose and Clatworthy (1955) showed that if there exists a PBIB design $D(v, b, r, k, \lambda_1 = 1, \lambda_2 = 0)$ based on a strongly regular graph $P(v, n_1, p_{11}^1, p_{11}^2)$ for which $r < k$, then parameters of D and P are given respectively by (1.3) and (1.4). In other words, the PBIB design has the same parameters as that arising from a partial geometry (r, k, t) . Bose (1963) proved later that the PBIB design is, in fact, a partial geometry (r, k, t) .

In this note we shall extend the result given above to arbitrary two-class PBIB designs, $\lambda_1 > \lambda_2$ and $r < k$, and show that these designs are special cases of partial geometric designs (r, k, t, c) first introduced by Bose, Shrikhande and Singhi (1976). We shall also express the parameters of any two-class PBIB design with $r < k$, and the parameters of the association scheme on which it is based, in terms of r, k, t, c, λ_2 . It will also be shown that such designs are precisely the special partially balanced incomplete block (SPBIB) designs studied by Bridges and Shrikhande (1974).

Partial geometric designs and SPBIB designs are defined in the next section.

2. Partial geometric and SPBIB designs

Consider a tactical configuration (v, b, r, k) for which the set of points is $V = \{x_1, x_2, \dots, x_v\}$ and the family of blocks is $\{B_1, B_2, \dots, B_b\}$ such that each point is contained in r blocks, and each block contains k points. We shall denote by $\lambda(x_i, x_\alpha)$, $i \neq \alpha$ the number of blocks containing both the points x_i and x_α and define $\lambda(x_i, x_i) = r$. Then

$$bk = vr, \quad \sum_{\alpha=1}^v \lambda(x_i, x_\alpha) = rk. \quad (2.1)$$

Let $N = (n_{ij})$ be the incidence matrix of the configuration. The configuration is said to be connected if N cannot be reduced to the form

$$N = \text{diag} \{N_1, N_2, \dots, N_g\}, \quad g > 1$$

by a permutation of the rows and columns, where N_i is a $v_i \times b_i$ matrix

$$\sum_{i=1}^g v_i = v, \quad \sum_{j=1}^g b_j = b.$$

Denoting the transpose of N by N^t we have

$$S = NN^t = (\lambda_{ia}). \quad (2.2)$$

If the configuration is connected, then rk is a simple eigenvalue of NN^t . Define

$$m(x_i; B_j) = \sum_{x_\alpha \in B_j} \lambda(x_i, x_\alpha) = \sum_{\alpha=1}^v \lambda(x_i, x_\alpha) n_{\alpha j}. \quad (2.3)$$

The configuration (v, b, r, k) is said to be a partial geometric design D with parameters (r, k, t, c) , $t \geq 1$, if for all i, j , $1 \leq i \leq v$, $1 \leq j \leq b$ we have

$$m(x_i, B_j) = \begin{cases} t & \text{if } x_i \notin B_j, \\ r + k - 1 + c & \text{if } x_i \in B_j. \end{cases} \quad (2.4)$$

We shall state here as lemmas certain results proved by Bose, Bridges and Shrikhande (1976).

Lemma 2.1. *The necessary and sufficient condition that a (v, b, r, k) configuration with incidence matrix N is a partial geometric design (r, k, t, c) is that*

$$NN^tN = tJ + (r + k + c - 1 - t)N. \quad (2.5)$$

The result (2.5) is equivalent to (2.4) and may be taken as an alternative definition of a partial geometric design (r, k, t, c) . It is easily seen that a partial geometric design (r, k, t, c) is a partial geometry (r, k, t) if $c = 0$.

Lemma 2.2. *If N is the incidence matrix of a partial geometric design with parameters (r, k, t, c) then NN^t has only a single non-zero eigenvalue*

$$\theta = r + k + c - 1 - t \quad (2.6)$$

other than the simple eigenvalue rk . The multiplicity of θ is given by

$$\alpha = \frac{rk[(r-1)(k-1)-c]}{t(r+k-t-1+c)}. \quad (2.7)$$

Lemma 2.3. *A connected configuration (v, b, r, k) with incidence matrix N is a partial geometric design (r, k, t, c) if and only if NN^t has a single non-zero eigenvalue θ other than the simple eigenvalue rk .*

It follows that a necessary condition for the existence of a partial geometric design (r, k, t, c) is that α , given by (2.7) is a non-negative integer.

A two-class PBIB design is called a special PBIB (SPBIB) design of the type $(\bar{s}, \bar{t})$ if there exist constants $\bar{s}$ and $\bar{t}$ such that for any point-block pair (x_i, B_j) the number of first associates of x_i in the block B_j is $\bar{s}$ or $\bar{t}$ according as $x_i \in B_j$ or $x_i \notin B_j$, i.e., according as $n_{ij} = 1$ or 0. As mentioned earlier these designs were first studied by Bridges and Shrikhande (1974).

3. Two-class PBIB designs for which $r < k$

Consider a two-class connected PBIB design $D(v, b, r, k, \lambda_1, \lambda_2)$ based on the association scheme $P(v, n_1, p_{11}^1, p_{11}^2)$ for which $r < k$ and $\lambda_1 > \lambda_2$. Let N be the incidence matrix of D . Then Bose and Mesner (1959) have shown that NN^t has just three eigenvalues, the simple eigenvalue rk and two other eigenvalues θ_1 and θ_2 given by

$$\theta_1 = r - \frac{1}{2}[(\lambda_1 - \lambda_2)(-\gamma - \sqrt{\Delta}) + (\lambda_1 + \lambda_2)], \quad (3.1)$$

$$\theta_2 = r - \frac{1}{2}[(\lambda_1 - \lambda_2)(-\gamma + \sqrt{\Delta}) + (\lambda_1 + \lambda_2)] \quad (3.2)$$

where

$$\gamma = p_{11}^1 - p_{11}^2 + 1, \quad \beta = 2n_1 - p_{11}^1 - p_{11}^2 - 1, \quad \Delta = \gamma^2 + 2\beta + 1. \quad (3.3)$$

If NN^t which is a $v \times v$ matrix has a zero eigenvalue, then the eigenvalue other than 0 and rk must be

$$\theta = \theta_1 + \theta_2 = 2r + \gamma(\lambda_1 - \lambda_2) - (\lambda_1 + \lambda_2). \quad (3.4)$$

It follows from Lemmas 2.2 and 2.3 that D is a partial geometric design (r, k, t, c) where θ is given by (2.6), and the multiplicity of θ is given by (2.7). Comparing (2.6) and (3.4) we have

$$k + c - 1 - t = r + \gamma(\lambda_1 - \lambda_2) - (\lambda_1 + \lambda_2). \quad (3.5)$$

If $r < k$, then $b < v$. Hence

$$\text{Rank } NN^t = \text{Rank } N \leq b < v.$$

It follows that NN^t has a zero eigenvalue and therefore D is a partial geometric design (r, k, t, c) for suitably chosen t and c .

Now let x_i be a point and B_j a block of D . If D is a partial geometric design (r, k, t, c) then from (2.3) and (2.4) if $x_i \notin B_j$ we have

$$\sum_{x_\alpha \in B_j} \lambda(x_i, x_\alpha) = t.$$

If $\bar{r}$ points of B_j are first associates of x_i , then the remaining $k - \bar{r}$ points of B_j are second associates of x_i . It follows from the definition of $\lambda(x_i, x_\alpha)$ that

$$t = \lambda_1 \bar{r} + (k - \bar{r})\lambda_2 = (\lambda_1 - \lambda_2)\bar{r} + k\lambda_2. \quad (3.6)$$

Therefore

$$\bar{r} = (t - k\lambda_2)/(\lambda_1 - \lambda_2). \quad (3.7)$$

Since the design is connected $\bar{r} \geq 1$. Since $\lambda_1 > \lambda_2$ from (3.6) we get $t > 1$.

Again if $x_i \in B_j$ then from (2.3) and (2.4)

$$\sum_{x_\alpha \in B_j} \lambda(x_i, x_\alpha) = r + k - 1 + c.$$

If $\bar{s}$ points B_j are first associates of x_i , then $k - 1 - \bar{s}$ points B_j are second associates of x_i . Also $\lambda(x_i, x_i) = r$. Hence we have

$$\begin{aligned} k - 1 + c &= \lambda_1 \bar{s} + \lambda_2 (k - 1 - \bar{s}) \\ c &= (\lambda_1 - \lambda_2)\bar{s} + (k - 1)(\lambda_2 - 1). \end{aligned} \quad (3.8)$$

Hence

$$\bar{s} = \{c - (k - 1)(\lambda_2 - 1)\}/(\lambda_1 - \lambda_2). \quad (3.9)$$

Thus $\bar{s}$ and $\bar{r}$ do not depend on the point block pair (x_i, B_j) . Hence from definition D is a SPBIB design of type $(\bar{s}, \bar{r})$.

We next express $\bar{s}$, $\bar{r}$ in terms of the parameters of the design D .

Let x_i be a fixed point of D . Let us count the number of point block pairs (x_α, B_j) such that B_j contains x_i , and x_α is a first associate of x_i contained in B_j . Then B_j can be chosen in r ways and corresponding to each choice of B_j there are $\bar{s}$ choices of x_α . Hence (x_α, B_j) can be chosen in $r\bar{s}$ ways. Again x_α can be chosen in n_1 ways and corresponding to each choice of x_α there are λ_1 choices of B_j . Hence

$$r\bar{s} = n_1 \lambda_1. \quad (3.10)$$

Similarly, counting the number of point block pairs (x_α, B_j) such that B_j does not contain x_i , and x_α is a first associate of x_i contained in B_j we have

$$(b - r)\bar{r} = n_1(r - \lambda_1). \quad (3.11)$$

since $b/k = v/r$ we have

$$r(v - k)\bar{r} = n_1 k(r - \lambda_1). \quad (3.12)$$

From (3.6) and (3.8) we then have

$$t = \frac{n_1 k(r - \lambda_1)(\lambda_1 - \lambda_2)}{r(v - k)} + k\lambda_2, \quad (3.13)$$

$$c = \frac{n_1 \lambda_1(\lambda_1 - \lambda_2)}{r} + (k - 1)(\lambda_2 - 1). \quad (3.14)$$

Hence we have

Theorem 3.1. *Given a two-class PBIB design D with parameters $(v, b, r, k, \lambda_1, \lambda_2)$ based on the association scheme (strongly regular graph) P with parameters $(v, n_1, p_{11}^1, p_{11}^2)$, and $r < k$, then D is a partial geometric design (r, k, t, c) where t and c are given by (3.13) and (3.14). D is also a SPBIB design of the type $(\bar{s}, \bar{r})$ where $\bar{s}$ and $\bar{r}$ are given by (3.10) and (3.12). Also t and $\bar{r}$ are connected by the relation (3.6) and c and $\bar{s}$ are connected by the relation (3.8).*

4. To express the parameters of D and P in terms of r, k, t, c and λ_2

Let D and P be as in Theorem 3.1, so that NN^t has a zero eigenvalue. We have the relations

$$NJ_{b,v} = rJ, \quad J_{b,v}N = kJ_{v,b}, \quad PJ_{v,b} = n_1J_{v,b}, \quad (4.1)$$

$$NN^t = (r - \lambda_2)I + gP + \lambda_2J, \quad (4.2)$$

where $g = \lambda_1 - \lambda_2$, and $J = J_{v,v}$. Again

$$PN = (\bar{s} - \bar{r})N + tJ_{v,b} \quad (4.3)$$

since

$$\sum p_{i\alpha} n_{\alpha j} = (1 - n_{ij})\bar{r} + n_{ij}\bar{s},$$

where $P = (p_{i\alpha})$.

Multiplying both sides of (4.3) by $J_{b,v}$ from the left and noting that $bk = vr$ we obtain

$$v\bar{r} = k(n_1 - \bar{s} + \bar{r}). \quad (4.4)$$

Again multiplying (4.2) from the right by $J_{v,b}$ we have

$$n_1 g = r(k - 1) - \lambda_2(v - 1). \quad (4.5)$$

Eliminating n_1 from (4.4) and (4.5) and using (3.7) and (3.9) we get

$$v = k[(r - 1)(k - 1) + t - c]/t, \quad (4.6)$$

$$b = r[(r - 1)(k - 1) + t - c]/t. \quad (4.7)$$

The expressions for v and b were obtained otherwise by Bose, Shrikhande and Singhi (1976).

Again from (3.10), (3.9) and (4.5) we have

$$\lambda_1 = \frac{r[c - (\lambda_2 - 1)(k - 1)]}{r(k - 1) - \lambda_2(v - 1)} \quad (4.8)$$

Substituting for v from (4.6) we get

$$\lambda_1 = \frac{rt[c - (\lambda_2 - 1)(k - 1)]}{t(r - \lambda_2)(k - 1) - \lambda_2 k[(r - 1)(k - 1) - c]} \quad (4.9)$$

this implies

$$g = \lambda_1 - \lambda_2 = \frac{\lambda_2^2[k(r - 1)(k - 1) - kc + t(k - 1)] - 2\lambda_2 tr(k - 1) + rt(c + k - 1)}{t(r - \lambda_2)(k - 1) - \lambda_2 k[(r - 1)(k - 1) - c]} \quad (4.10)$$

Then from (4.5) and (4.6) we have

$$n_1 g = (r - \lambda_2)(k - 1) - \frac{\lambda_2 k}{t}[(\lambda - 1)(k - 1) - c] \quad (4.11)$$

Again [Bose and Mesner (1959)]

$$P^2 = n_1 I + p_{11}^1 P + p_{11}^2 (J - I - P). \quad (4.12)$$

Hence

$$P^2 N = (n_1 - p_{11}^2)N + (p_{11}^1 - p_{11}^2)PN + p_{11}^2 kJ. \quad (4.13)$$

Using (4.3) we get

$$P^2 N = P(PN) = (\bar{s} - \bar{r})^2 N + \bar{r}(\bar{s} - \bar{r} + n_1)J. \quad (4.14)$$

The above two equations give

$$(\bar{s} - \bar{r})p_{11}^1 - (\bar{s} - \bar{r} + 1)p_{11}^2 = (\bar{s} - \bar{r})^2 - n_1, \quad (4.15)$$

$$\bar{r}p_{11}^1 + (k - \bar{r})p_{11}^2 = \bar{r}(\bar{s} - \bar{r} + n_1). \quad (4.16)$$

Solving (4.14) and (4.15) we get after some calculations

$$p_{11}^1 = \frac{(\bar{s} - \bar{r})[k(\bar{s} - \bar{r}) + \bar{r}(n_1 + 1)] + n_1(2\bar{r} - k)}{[k(\bar{s} - \bar{r}) + \bar{r}]} \quad (4.17)$$

and

$$p_{11}^2 = \frac{\bar{r}n_1(\bar{s} - \bar{r} + 1)}{[k(\bar{s} - \bar{r}) + \bar{r}]} \quad (4.18)$$

From (3.7) and (3.9) we now have

$$p_{11}^1 = \frac{(k+c-t+\lambda_2-1)[k(k+c-t-1)+t+n_1(t-\lambda_2k)]+n_1[2(t-\lambda_2k)-kg]}{g[k(k+c-t-1)+t]}, \quad (4.19)$$

$$p_{11}^2 = \frac{n_1(t-\lambda_2k)[c+k-1-t+g+\lambda_2]}{g[k(k+c-t-1)+t]}. \quad (4.20)$$

Finally substituting for n_1 and g from (4.10) and (4.11) we obtain p_{11}^1 and p_{11}^2 in terms of r, k, t, c and λ_2 . We thus have

Theorem 4.1. *The parameters of the two-class PBIB design D of Theorem 3.1 and its association scheme P can be expressed in terms of r, k, t, c and λ_2 . These expressions are given by the equations (4.6)–(4.11), (4.19) and (4.20).*

If in particular $\lambda_1 = 1, \lambda_2 = 0$, then (4.9) gives $rc = 0$. Since $t \geq 1$, we have $c = 0$. Hence the design D becomes a partial geometry (r, k, t) . We can then check using (4.6), (4.7), (4.10), (4.11), (4.19) and (4.20) that the values of $v, b, r, k, n_1, p_{11}^1$ and p_{11}^2 are as in (1.3) and (1.4).

References

- [1] Bose, R.C. (1963). Strongly regular graphs, partial geometries and partially balanced designs. *Pacific J. Math.* 13, 389–419.
- [2] Bose, R.C., W.G. Bridges and M.S. Shrikhande (1976). A characterization of partial geometric designs. *Discrete Math.* 16, 1–7.
- [3] Bose, R.C. and W.H. Clatworthy (1955). Some classes of partially balanced designs. *Ann. Math. Statist.* 26, 212–232.
- [4] Bose, R.C. and W.S. Connor (1952). Combinatorial properties of group divisible designs. *Ann. Math. Statist.* 23, 367–383.
- [5] Bose, R.C. and D. Mesner (1959). On linear associative algebras corresponding to association schemes of partially balanced designs. *Ann. Math. Statist.* 30, 21–33.
- [6] Bose, R.C. and K.R. Nair (1939). Partially balanced incomplete block designs. *Sankhya* 4, 337–372.
- [7] Bose, R.C. and T. Shimamoto (1952). Classification and analysis of partially balanced incomplete block designs with two associate classes. *J. Amer. Statist. Assoc.* 47, 151–184.
- [8] Bose, R.C., S.S. Shrikhande and K.N. Bhattacharya (1953). On the construction of group divisible designs. *Ann. Math. Statist.* 24, 167–195.
- [9] Bose, R.C., S.S. Shrikhande and N.M. Singhi (1976). Edge regular multigraphs and partial geometric designs with an application to the embedding of quasi-residual designs. *Proceedings of the International Colloquium on Combinatorial Theory (Rome, Sept. 1973)*, Vol. 1, 49–81.
- [10] Bridges, W.G. and M.S. Shrikhande (1974). Special partially balanced incomplete block designs and associated graphs. *Discrete Math.* 9, 1–18.
- [11] Dembowski, P. (1968). *Finite Geometries*, Springer, Berlin, Heidelberg, New York.
- [12] Nair, K.R. and C.R. Rao (1942). A note on partially balanced incomplete block designs. *Science and Culture* 7, 568–569.